

Pattern:

A pattern is an arrangement of repeating numbers or shapes.



The recurring characteristic in this pattern is the colour blue, red. This repeating element indicates the pattern's order.

Extended Pattern:

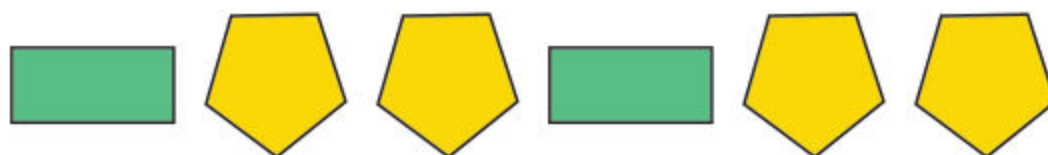
The pattern in this example starts with a circle, followed by a triangle, and a pentagon. As the shapes are repeating in this pattern, the next two shapes after the pentagon are the circle and the triangle.

**Example 1**

Here is a pattern made with 3 D shapes. The pattern in this example starts with a cone followed by a cube, and a cuboid.

**Exercise 1**

1. What shape will appear in the following pattern of shapes?

**Solution:**

From the image, you can observe that the pattern consists of a rectangle followed by two pentagons. So in this sequence, the rectangle will be repeated, followed by two pentagons.

Unit 5: Algebra (Patterns)

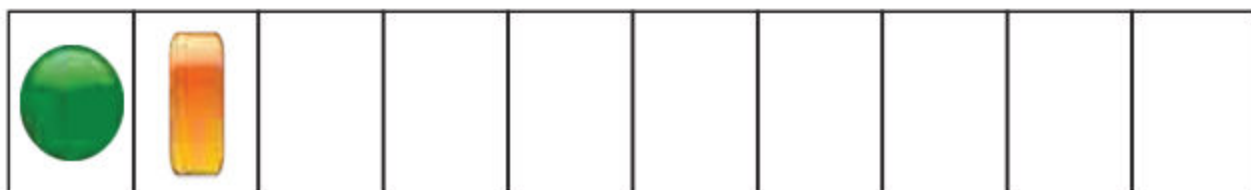
2. Find the missing emoji in the following pattern.



3. What follows next, the pizza or the cake?



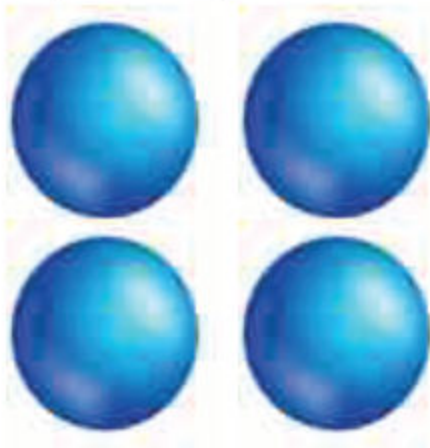
4. Extend the patterns



5. Match the pattern in Column A with the same pattern in Column B.



6. You have spheres, cones, cubes and cuboids, four sets of each 3D shape. What patterns could you make?



Unit 5: Algebra (Patterns)

Patterns of a Hundred Chart:

Here is a 100-point chart.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

There are several techniques to locate the pattern. For instance,

For each row, left to right, the number increases by 1.

For each column, from top to bottom, the number increases by 10.



Activity

Make a pattern by adding plus two with the initial number and choosing any number as a starting point.

Example:

- i) 2, 4, 6, 8,
- ii) 23, 25, 27, 29,

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Exercise 2

1. Using hundred-point chart, make a pattern by adding plus three with the initial number.
2. Using hundred-point chart, make a pattern by adding plus four with the initial number.
3. What would be the next number in the given number sequences:
 - a) 41, 45, 49,
 - b) 34, 44, 54,

MEASUREMENT

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ recognize and use the standard units of length (meter and centimeter) to measure and record the length of different objects.
- ✓ compare the length of different objects using standard units of length (meter and centimeter) using greater than, less than and equal to.
- ✓ add and subtract lengths, given in the same units to solve real-life word problems.
- ✓ recognize and use the standard units of mass (kilograms and grams) to measure and record the mass of different objects.
- ✓ compare the mass of different objects using standard units of mass (kilogram and gram) using greater than, less than and equal to.
- ✓ add and subtract mass given in the same units to solve real-life word problems.
- ✓ recognize and use the standard units of capacity (liter and milliliter) to measure and record the capacity of different objects.
- ✓ compare the capacity of different objects using standard units of capacity (liter and milliliter) using greater than, less than and equal to.
- ✓ add and subtract capacities given in the same units to solve real-life word problems.
- ✓ read and write temperature to the nearest appropriate unit i.e. ($^{\circ}\text{C}$) using pictorial representation and relating temperature scale to number line.